

The **SUMIF** function is a mathematical conditional function used to add values in a specific range that meet a single, defined criterion.

1. Core Syntax

The function operates using three main components:

`=SUMIF(range, criteria, [sum_range])`

- **range:** The group of cells you want to evaluate based on your criteria.
- **criteria:** The condition that determines which cells to add (e.g., ">1000", "Hubli", or a specific date).
- **sum_range** (Optional): The actual cells to add if they are different from the evaluation range. If omitted, Excel adds the cells in the first `range` argument.

2. Practical Examples

Using the concepts found in typical data management tasks, here is how the function is applied:

- **Text Criteria:** To calculate the total salary for a specific city, the formula would look like: `=SUMIF(E2:E11, "Hubli", D2:D11)`.
- **Numerical Criteria:** To find the total of all scores above a passing mark, you might use: `=SUMIF(F2:F11, ">35")`.
- **Cell References:** Instead of typing "Hubli," you can reference a cell (e.g., `=SUMIF(E2:E11, H1, D2:D11)`) to make the calculation dynamic.

3. Key Features for Engineering Workflows

- **Logic Integration:** SUMIF uses the same logical operators as the **IF function** (e.g., `=`, `>`, `<>`) to filter which data is included in the sum.
- **Data Integrity:** For SUMIF to work accurately, data should be standardized using tools like **Data Validation** to prevent typos from excluding relevant values.
- **Automatic Updating:** If you change a value in the range, the SUMIF result updates immediately, providing real-time data analysis.